

Myungkyu Lee

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RESEARCH INTERESTS

Symmetric-Key Cryptography, Cryptanalysis, Geometric Approach, Quasidifferential Framework

EDUCATION

- **Korea University, School of Cybersecurity** Mar. 2025 – Expected Feb. 2027
M.S. in Information Security, Cryptographic Algorithm Laboratory (Advisor: Prof. Seokhie Hong) Seoul, Korea
◦ GPA: 4.50/4.50 (100/100; coursework completed)
- **University of Seoul** Mar. 2020 – Feb. 2025
B.S. in Mathematics; Double Major in Computer Science Seoul, Korea
◦ GPA: 4.38/4.50 (98.8/100), Rank: 2/29

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, U=UNDER REVIEW, E=EPRINT PREPRINT, T=THESIS, D=DOMESTIC (KOREAN)

- [U.1] **Myungkyu Lee**, Yunjae Hwang, Hanbeom Shin, Insung Kim, Sunyeop Kim, Byoungjin Seok, Dongjae Lee, Deukjo Hong, Jaechul Sung, Seokhie Hong (2026). **Revisiting Differential Characteristics of SPECK and CHAM via Extended Quasidifferential Framework**. *Under review*.
- [E.2] **Myungkyu Lee**, Byoungjin Seok, Dongjae Lee, Deukjo Hong, Jaechul Sung, Seokhie Hong (2026). **Rotational-Quasidifferential Framework – A Geometric Approach to Rotational-XOR Cryptanalysis**. *Cryptology ePrint Archive*, Paper 2026/1633.
- [E.1] Yunjae Hwang, Insung Kim, Sunyeop Kim, **Myungkyu Lee**, Hanbeom Shin, Deukjo Hong, Seokhie Hong, Dongjae Lee, Jaechul Sung, Byoungjin Seok (2026). **Improving Neural-Inspired Integral Distinguishers via a Linear-Algebraic Approach**. *Cryptology ePrint Archive*, Paper 2026/340.
- [D.5] **Myungkyu Lee**, Hanbeom Shin, Insung Kim, Sunyeop Kim, Donggeun Kwon, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Hash Table Method for Data Search in Differential Cryptanalysis**. *Journal of The Korea Institute of Information Security and Cryptology (JKIISC)*, Dec. 2025.
- [D.4] Yunjae Hwang, **Myungkyu Lee**, Insung Kim, Hanbeom Shin, Sunyeop Kim, Byoungjin Seok, Seokhie Hong, Sangjin Lee (2025). **Integral Cryptanalysis on SKINNY-64-64 using PCA**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2025 (CISC-W '25)*, Nov. 2025.
- [D.3] **Myungkyu Lee**, Yunjae Hwang, Hanbeom Shin, Insung Kim, Sunyeop Kim, Byoungjin Seok, Dongjae Lee, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Security Analysis of Alzette in the Fixed-Key Model**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2025 (CISC-W '25)*, Nov. 2025.
- [D.2] **Myungkyu Lee**, Hanbeom Shin, Insung Kim, Sunyeop Kim, Donggeun Kwon, Deukjo Hong, Jaechul Sung, Seokhie Hong (2025). **Efficient Pair Search Techniques for Zero Difference-Based AES Cryptanalysis**. In *Proceedings of the Conference on Information Security and Cryptography Summer 2025 (CISC-S '25)*, Jun. 2025.
- [D.1] **Myungkyu Lee**, Jaechul Sung, Seokhie Hong (2024). **Randomness Testing on Block Cipher GIFT-128**. In *Proceedings of the Conference on Information Security and Cryptography Winter 2024 (CISC-W '24)*, Nov. 2024.

AWARDS AND HONORS

- **Grand Prize (1st prize)** Fall 2025
National Cryptography Expert Training (11th Cohort), NSR & Korea Cryptography Forum
- **Special Award** Fall 2025
Cryptographic Analysis Contest, Korea Institute of Information Security and Cryptology (KIISC)
- **Grand Prize (1st prize)** Fall 2024
University Algorithm Competition (UOSPC) (Division 1), University of Seoul
- **Silver Medal (Field 1)** Fall 2024
Undergraduate Mathematics Competition, Korean Mathematical Society (KMS)
- **Special Award** Fall 2024
Cryptographic Analysis Contest, Korea Institute of Information Security and Cryptology (KIISC)
- **Grand Prize (1st prize)** Summer 2024
Gyeonggi University of Science and Technology Summer Big Data Coding Competition, Gyeonggi Univ. of Science & Tech.
- **AI Convergence Innovation Division Director's Award (3rd Prize)** Fall 2023
Industry-Academia Deep Learning Winter Bootcamp Competition, University of Seoul

• Participation Prize (3rd prize) <i>AICOSS Hackathon, University of Seoul</i>	Fall 2023
• Participation Prize (3rd prize) <i>University Algorithm Competition (UOSPC) (Division 2), University of Seoul</i>	Fall 2023
• Participation Prize (3rd prize) <i>Creative Engineering Mathematics Competition, University of Seoul</i>	Fall 2023
• Participation Prize (3rd prize) <i>Gyeonggi University of Science and Technology Summer Big Data Coding Competition, Gyeonggi Univ. of Science & Tech.</i>	Summer 2023
• Grand Prize (1st prize) <i>University Algorithm Competition (UOSPC) (Division 2), University of Seoul</i>	Fall 2021
• Excellence Award (2nd prize) <i>Creative Engineering Mathematics Competition, University of Seoul</i>	Fall 2021
• Excellence Award (2nd prize) <i>Creative Engineering Mathematics Competition, University of Seoul</i>	Fall 2020

SCHOLARSHIPS

• Scholarship under the University Innovation Support Program <i>University of Seoul (Division of Academic Affairs)</i>	Fall 2024
• National Scholarship for Science and Engineering <i>Ministry of Science and ICT, Republic of Korea</i>	Fall 2024
• National Scholarship for Science and Engineering <i>Ministry of Science and ICT, Republic of Korea</i>	Spring 2024
• Extracurricular Education Scholarship <i>University of Seoul</i>	Spring 2024
• Scholarship under the Innovation Sharing University Project <i>University of Seoul (Sponsored by UOS Industry Cooperation Foundation)</i>	Fall 2023
• Scholarship for Excellent Achievement <i>University of Seoul</i>	Spring 2022
• Scholarship for Excellent Achievement <i>University of Seoul</i>	Fall 2021
• Seoul Mayor's Scholarship (Top Honor) <i>University of Seoul</i>	Spring 2021
• Scholarship for Excellent Achievement <i>University of Seoul</i>	Fall 2020

TEACHING AND MENTORING EXPERIENCE

• SHA (UOS Hacking Club) — Cryptography Study Group <i>Study Leader; Mentor</i>	Fall 2024
• Elementary Linear Algebra <i>Teaching Assistant (Paid Position, Lecturer: Prof. Jeong-Ah Kim)</i>	Fall 2024
• Mathematics and Computer <i>Teaching Assistant (Paid Position, Lecturer: Prof. Jaechul Sung)</i>	Fall 2024
• PseudoCode (Algorithm Club in the Department of Mathematics, UOS) <i>Founder & Club Leader</i>	Summer – Winter 2024
• Discrete Mathematics <i>Teaching Assistant (Paid Position, Lecturer: Prof. Jaechul Sung)</i>	Spring 2024
• Calculus II <i>Course Tutor (Paid Position)</i>	Fall 2021

ADDITIONAL ACADEMIC TRAINING

• National Cryptography Expert Training (11th Cohort) <i>Selected Trainee (15 selected nationwide); Organized by NSR & Korea Cryptography Forum</i>	May – September 2025
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MILITARY SERVICE

• Republic of Korea Army <i>Compulsory Military Service as a C4I Systems Operator and Maintainer</i>	Jan. 2022 – Jul. 2023
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TECHNICAL SKILLS

- **Programming Languages:** Python, C, C++ (Advanced), SageMath (Basic)
- **Frameworks/Tools:** PyTorch, TensorFlow, Flask (Intermediate)

ADDITIONAL INFORMATION

Languages: Korean (native), English (advanced)
Interests: Squash, Badminton, and Calisthenics